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CHANGE RECORD

When the quality of the products changes, the QuID is updated and the SQO is updated. A line is added to this table and the version of the SQO document is the same than that of the REFERENCE QUID. The third column specifies which sections or sub-sections have been updated.

Issue	Date	§	Description of Change	Authors	Validated By
1.4	01/07/23	All	Creation of the document	Stéphane LAW CHUNE	Validated by PQ leader
1.5	31/05/24		Update QUID version	S. LAW CHUNE	O. Le Galloudec
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Executive summary

The quality of the GLOBAL_MULTIYEAR_WAV_001_032 has been assessed by comparison with observations and insitu data to describe the space-time evolution of the following variables:

* Significant wave height (SWH)
* Peak wave period
* Mean wave Period/ Zero up-crossing period
* Mean wave direction

GLOBAL_MULTIYEAR_WAV_001_032 for the global wave reanalysis describing past sea states over the years 1993-2022. This product also bears the name of WAVERYS and as such it will be referred hereafter.

The quality of the WAVERYS wave reanalysis was estimated by: (i) comparing the SWH analyzed with those measured by the altimeter of the HY-2A satellite (non-assimilated data, open ocean, validation over 2014-2018); (ii) comparing the mean wave parameters with CMEMS in-situ buoy measurements (coastal data, validation over 1993-2015); and finally, by (iii) comparing the time variations of these mean parameters with those of the ERA5 wave reanalysis over the whole series. The quality assessment covers a subset of key wave parameters, results of which are summarised below.

The core of WAVERYS is based on the MFWAM model, a third-generation wave model that calculates the wave spectrum, i.e., the distribution of sea state energy in frequency and direction on a 1/5° irregular grid. The wave parameters derived from the wave spectrum, such as significant wave height (SWH) or the mean wave period, are delivered globally on a regular grid size of 1/5° with a 3-hour time step. The wave spectrum is discretized into 30 frequencies obtained from a geometric sequence of first member 0.035 Hz and a common ratio of 7.5. WAVERYS considers oceanic currents from the GLORYS12 physical ocean reanalysis and assimilates significant wave height observed from historical altimetry missions and directional wave spectra from Sentinel 1 SAR from 2016 onwards.

For additional information regarding the in-depth validation of this product, the calculation of the assessment metrics presented in this product other detailed information in quality and noticeable events please refer to the reference quid document CMEMS-GLO-QUID-001-032 and the PUM.

Important notice:
The contents of this document are an assessment based on the best set of observations available for evaluation at the time the operational system was validated. The validation methodology was defined and agreed within CMEMS, inheriting the long experience of MyOcean and MERSEA series of projects (Hernandez et al., 2018) The results presented in this report and derived estimated accuracy numbers (EAN) are representative of average error levels over large areas of the ocean. These numbers might be used as a mean error in one given point of the area, but in order to refine error estimates locally, the reader is invited to use complementary information from the reference QUIDs (error maps for instance, when available).

1. Significant Wave Height

Significant wave height (SWH): For the open sea, WAVERYS performs with a global scatter index of 8.5% with relatively low biases about 3 cm. The quality of the analyses is better for tropical latitudes than for high latitudes where the scatter index reaches more than 10%. This is particularly the case for coastal areas and/or those subject to complex oceanic processes (e. g. instability of currents, tides, etc.), where the scatter index can reach 20%. Generally, the wave model performs worse in light wind conditions (typically summer) than in strong wind conditions that effectively constrain the model (typically winter), with differences of a few percent on the scatter index.

Table 1: Estimated accuracy numbers corresponding to the comparison between model and in situ CMEMS observations for the region and the sub-regions

Variable	Metrics	Value
Open-Ocean Significant wave height	mean bias, RMS, SI	3 cm / 22 cm/ 8.8%
Coastal Significant wave height	mean bias, RMS, SI	-6 cm / 37 cm / 22.2%

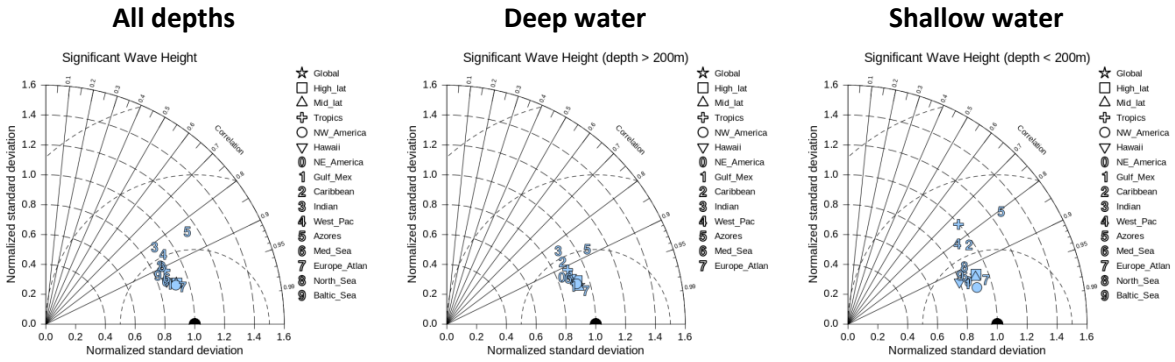


Figure 1: Normalized Taylor diagrams of SWH of WAVERYS in comparison with buoys. Statistics for different regions, all depths (top panel), depth deeper than 200 m (left panel), and shallower than 200 m (right panel).

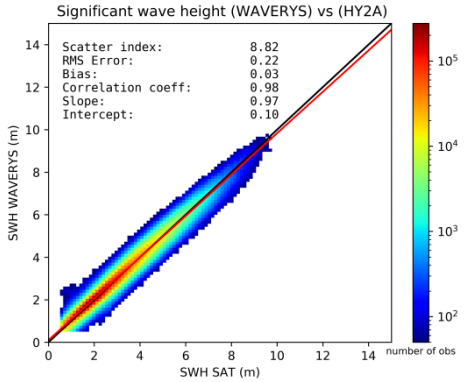


Figure 2: Scatter diagram between the SWH measured with HY-2A and the model WAVERYS. The colour shade represents the observation density. Statistical scores are given, as well as the slope and y-axis of the associated linear regression.

2. Peak Wave Period

Validation of the following quantities is only available for a limited set of coastal moorings located mainly in the Atlantic and Pacific of the Northern Hemisphere:

Peak wave period: Despite the difficulty of validation in coastal regime, the bias in peak periods are about 10 milliseconds (ms). There is an area identified along the Iberian coast and in the Azores with scatter index around 40% (compared to 26% of SI for all the measurements).

Table 2: Estimated accuracy numbers corresponding to the comparison between model and in situ observations for the coastal area.

Variable	Metrics	Value
Coastal Peak wave period	mean bias, RMS, SI	0.16 s / 2.3 s / 26.4%

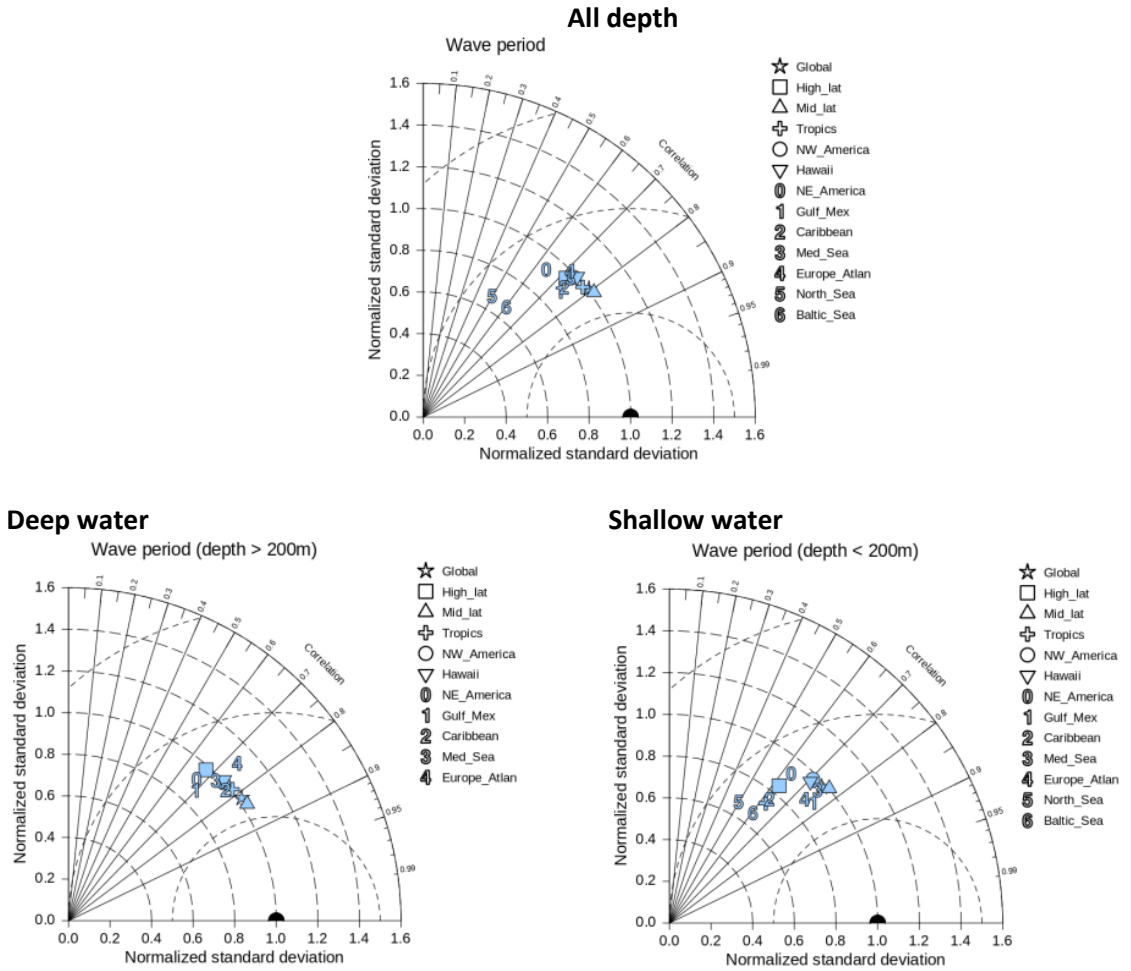


Figure 3: Normalized Taylor diagrams of PWP of WAVERYS in comparison with buoys. Statistics for different regions, all depths (top panel), depth deeper than 200 m (left panel), and shallower than 200 m (right panel).

3. Mean Wave Period

Validation of the following quantities is only available for a limited set of coastal moorings located mainly in the Atlantic and Pacific of the Northern Hemisphere:

Zero up-crossing period: We can note a bias of the mean wave period of about 40 ms, but with low index scatter indexes, of about 15%. WAVERYS is particularly efficient at capturing Pacific Ocean swells (scatter index of 11% in Hawaii).

Table 3: Estimated accuracy numbers corresponding to the comparison between model and in situ observations for the coastal area

Variable	Metrics	Value
Coastal Mean wave period	mean bias, RMS, SI	0.4 s / 0.87 s / 14.9%

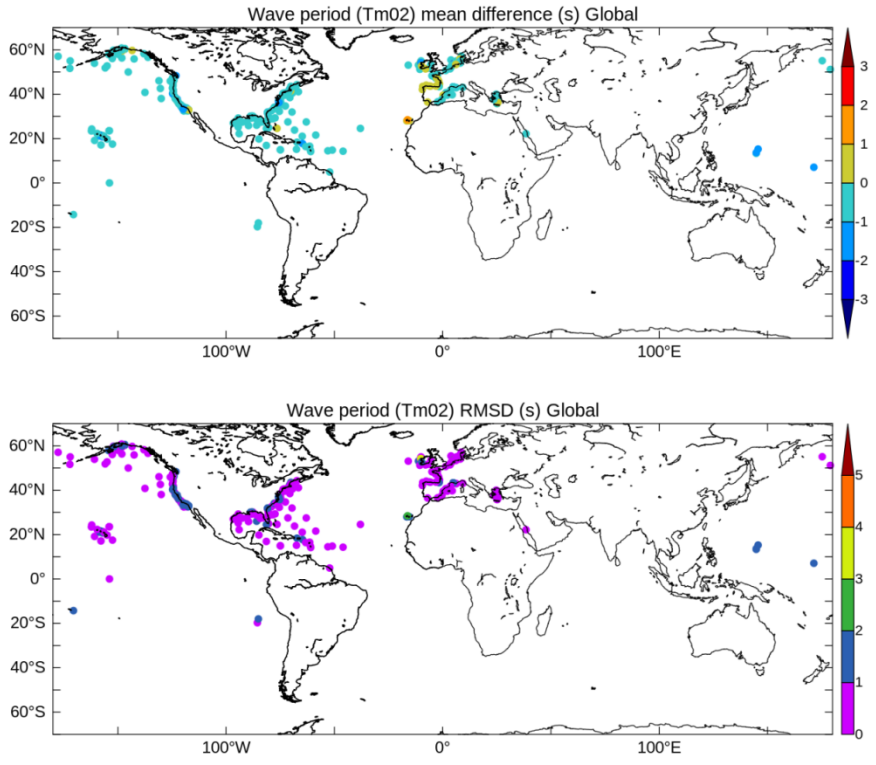


Figure 4: Bias (top) and RMSD (bottom) of mean wave period (sec) of WAVERYS in comparison with CMEMS buoys averaged over the observed period.

4. Mean Wave Direction

Validation of the following quantities is only available for a limited set of coastal moorings located mainly in the Atlantic and Pacific of the Northern Hemisphere:

Wave mean direction: The average direction deviation (bias) for all buoy measurements is about 5° for a value of 38° of RMSD. Smaller differences can be noted in the Tropics, where swell regimes prevail (differences of around 1°). Due to wave kinematics, the directions are better constrained at the coast than in the open ocean.

Table 4: Estimated accuracy numbers corresponding to the comparison between model and in situ observations for the coastal area.

Variable	Metrics	Value
Coastal Mean wave direction	mean bias, RMS	4.98 deg/ 37.8 deg

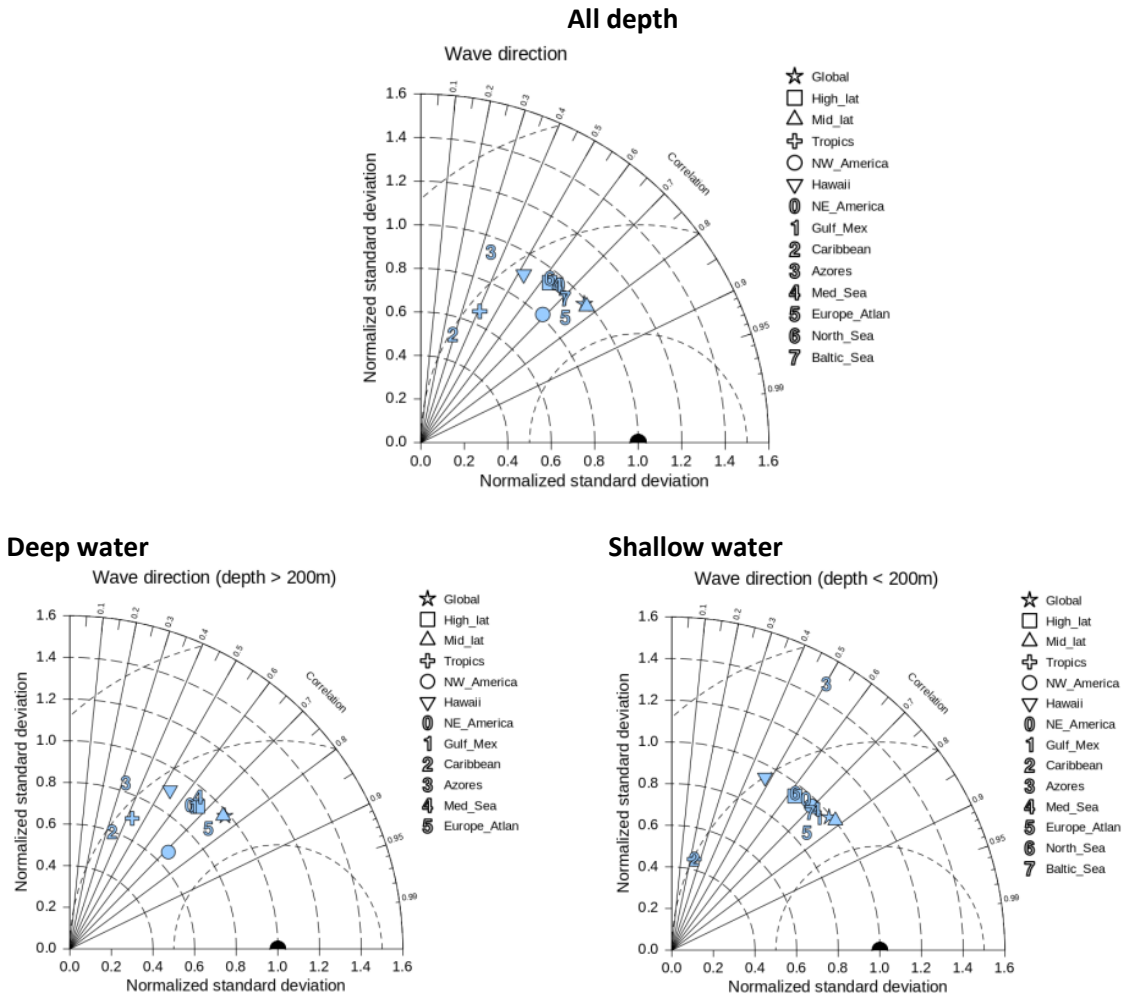


Figure 5: Normalized Taylor diagrams of MWD of WAVERYS in comparison with buoys. Statistics for different regions, all depths (top panel), depth deeper than 200 m (left panel), and shallower than 200 m (right panel).

References

Hernandez, F., et al., 2018: Measuring performances, skill and accuracy in operational oceanography: New challenges and approaches. In "New Frontiers in Operational Oceanography", E. Chassignet, A. Pascual, J. Tintoré, and J. Verron, Eds. GODAE OceanView, 759-796, doi:10.17125/gov2018.ch29.